C8.1 About barrier transactions

Barrier transactions provide guarantees about the ordering and observation of transactions in a system. ACE supports memory barriers and synchronization barriers:

- A memory barrier is issued by a master to guarantee that if another master in the appropriate domain can observe any transaction issued after the barrier it must be able to observe every transaction issued before the barrier.
- A synchronization barrier is issued by a master to guarantee that all transactions issued before the barrier are observable by every master in the appropriate domain when the barrier completes. System domain synchronization barriers have the additional requirement that all transactions issued before the barrier transaction must have reached the endpoint slaves they are destined for before the barrier completes.

A memory barrier is used for memory based communication. For example, when writing an array of data to memory, a master component can issue a memory barrier before setting a memory flag to indicate that the array is available. Any other master component that can observe the flag must observe all transactions that write to the array.

Note

It is not necessary for all master components in the domain to observe the updated array at the same time. It is a requirement for each master in the domain that can observe the flag, to be guaranteed to observe the updated array.

A synchronization barrier is used with various forms of sideband signaling communication. For example, when writing an array of data to memory, a master component can use a synchronization barrier before generating an interrupt to indicate that the array is available. When the synchronization barrier completes, the updated array is guaranteed to be observable by all master components in the domain.

Barrier transactions can be read or write transactions, and are defined as follows:

Read barrier transactions

A master component issues a read barrier transaction on the read address channel and a response is returned on the read data channel. No data transfer occurs.

Write barrier transactions

A master component issues a write barrier on the write address channel and a response is returned on the write response channel. No data transfer occurs.

A master component must issue barrier transactions as a barrier pair, with a barrier transaction on both the read address channel and the write address channel. For each address channel, any transaction issued on the channel, before the barrier transaction, is defined to be before the barrier, even if it is issued after the corresponding barrier on the other address channel. A transaction is defined to be after the barrier if it is issued after both the read barrier response and write barrier response are received.

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